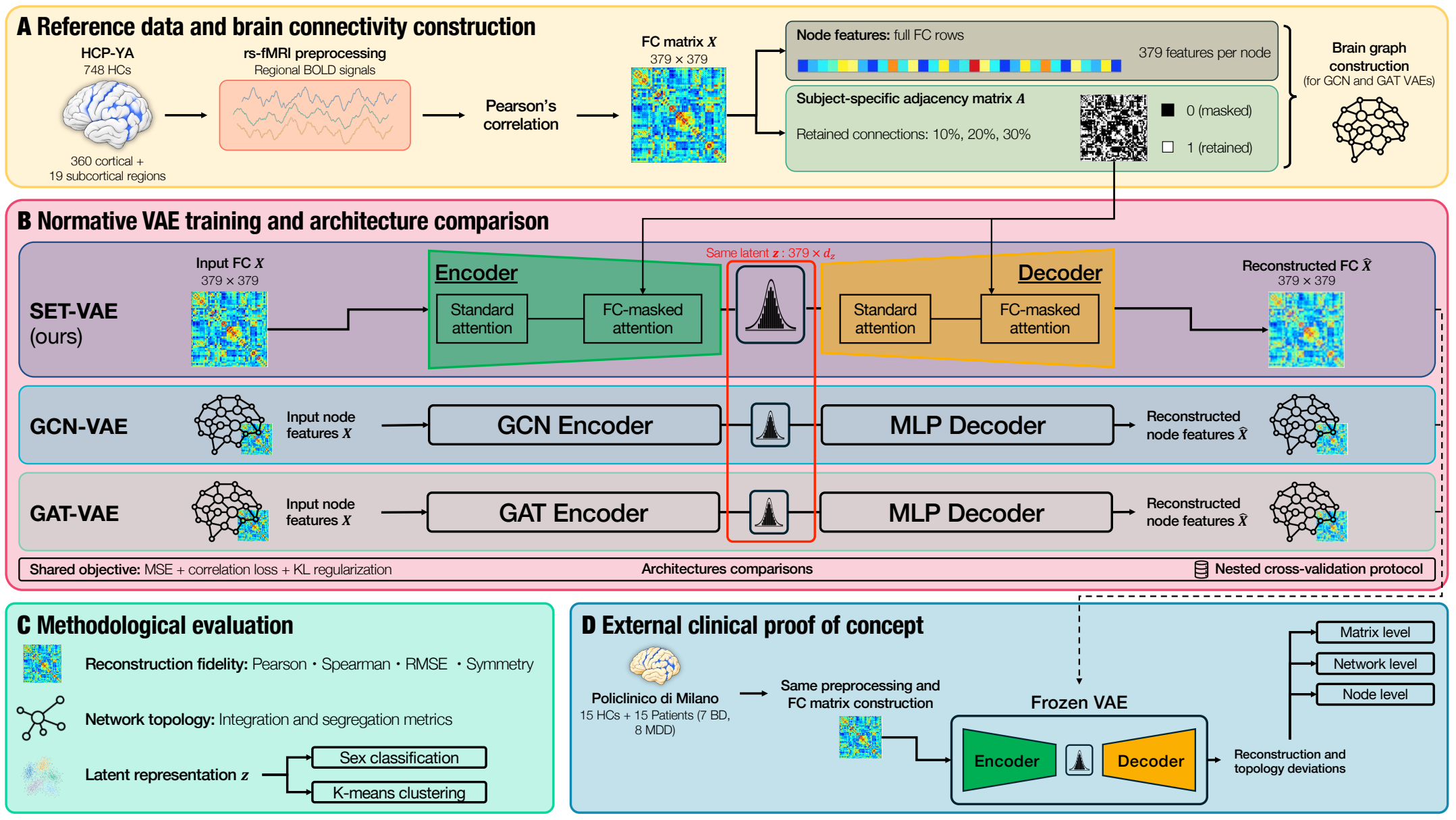

A Reference data and brain connectivity construction
HCP-YA
748 HCs
360 cortical +
19 subcortical regions
rs-fMRI preprocessing
Regional BOLD signals
Pearson's correlation
FC matrix X
379 × 379
Node features: full FC rows
379 features per node
Subject-specific adjacency matrix A
Retained connections: 10%, 20%, 30%
0 (masked)
1 (retained)
Brain graph construction
(for GCN and GAT VAEs)
B Normative VAE training and architecture comparison
SET-VAE
(ours)
Input FC X
379 × 379
Encoder
Standard attention
FC-masked attention
Same latent z : 379 × d_z
Decoder
Standard attention
FC-masked attention
Reconstructed FC X̂
379 × 379
GCN-VAE
Input node features X
GCN Encoder
MLP Decoder
Reconstructed node features X̂
GAT-VAE
Input node features X
GAT Encoder
MLP Decoder
Reconstructed node features X̂
Shared objective: MSE + correlation loss + KL regularization
Architectures comparisons
Nested cross-validation protocol
C Methodological evaluation
Reconstruction fidelity: Pearson • Spearman • RMSE • Symmetry
Network topology: Integration and segregation metrics
Latent representation z
Sex classification
K-means clustering
D External clinical proof of concept
Policlinico di Milano
15 HCs + 15 Patients (7 BD, 8 MDD)
Same preprocessing and FC matrix construction
Frozen VAE
Encoder
Decoder
Reconstruction and topology deviations
Matrix level
Network level
Node level